Launching Mexico's Space Endeavors: An Integrated Approach to the Geographic Location of a Mexican Spaceport and Legal Regulation of Propellants

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Abstract

In recent years, there has been a growing trend towards the development of satellites and constellations with missions in remote sensing and telecommunications operating in Low Earth Orbit. This research addresses the analysis of Mexican context in order to propose a geographic location of a spaceport taking into consideration classical orbital mechanics elements, as well as the legal requirements for handling, manufacturing and transporting the propellants for these rockets. A Mexican spaceport site would not only offer advantages for launches due to its relative proximity to the equator, but also to develop the Mexican space industry. Furthermore, meteorological, social, environmental and legal factors are studied, making this territory an ideal launch zone and analyzing the necessary legal framework with the purpose of enhancing the development of the space sector in Mexico.

1 Background

The development of aerospace technology has experienced exponential growth in recent decades. Advances made after World War II and during the Cold War led to a race for space dominance between the former socialist and capitalist blocs. Throughout the twentieth century, various launch sites were developed because of

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the Cold War between the United States of America (USA) and the Soviet Union, initially for military and governmental purposes. They were later used for scientific experiments and eventually for launching commercial satellites and building the International Space Station.

In Mexico, the first rocketry activities began in 1957 when engineer Walter C. Buchanan assembled a group of engineers from the Autonomous University of San Luis Potosi, dedicated to experimental rocketry, who designed, built, and launched the first liquid-fueled rockets. Subsequently, the Secretariat of Communications and Transportation developed the SCT1 and SCT2 rockets with the mission of conducting high-altitude measurements, with apogees of four and 25 km respectively; however, an orbital flight has never been achieved.[1]

In 1962, the National Commission of Outer Space (CONEE) was established by Presidential Decree with the aim of promoting research, exploration, and peaceful use of outer space. However, it was dissolved in 1973 due to the aftermath of the oil crisis.[2]

In 2010, the Mexican Space Agency was created through a law published in the Mexico Official Federation Gazette, establishing it as a Public Decentralized Organization with the purpose of promoting space activities in the aerospace sector. Mexico's Space Policy is defined through the General Guidelines of Mexico's Space Policy, which were proposed by the Governing Board of the Mexican Space Agency in 2011. These guidelines are implemented through the National Program of Space Activities.[3]

The Mexican Space Agency has the Orbit Plan 2.0, a roadmap for the Mexican space sector, which defines its strategies and challenges, demonstrating that Mexico is on track to develop more space activities and position itself as a leader in space in the Latin American region.[4]

On 30 March 2023, a constitutional reform to Articles 23 and 78 was approved in the Chamber of Deputies, prioritizing all space activities for national development, without limiting the constitutional framework to only consider satellite communication. It also allows the participation of non-governmental actors in

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[1] Hacia el Espacio, (2017), "*¿Sabías qué? - Cohetes y Sputnik*", (all websites cited in this publication were last accessed and verified on 14 July 2023), https://haciaelespacio.aem.gob.mx/revistadigital/articul.php?interior=592.

[2] Official Gazette of the Federation, 01, 63–207, "*National Space Program Activities 2024"*, Mexican Space Agency, www.dof.gob.mx/nota_detalle.php?codigo=5608451&fecha=22/12/2020#gsc.tab=0.

[3] House of Representatives of the Honorable Congress of the Union, (n.d.), "*Law of the Mexican Space Agency*", www.diputados.gob.mx/servicios/datorele/cmprtvs/iniciativas/Inic/339/2.html.

[4] Official Gazette of the Federation, 01, 63–207. "*National Space Activities Program 2024*", Mexican Space Agency, www.dof.gob.mx/nota_detalle.php?codigo=5608451&fecha=22/12/2020#gsc.tab=0.

space-related activities under the direction of the Mexican State, for the purpose of international responsibility. This marks the beginning of formalization and regulation of space activities.[5]

2 Introduction

Currently, there are up to 35 spaceports worldwide from which launches can be conducted for various missions and objectives. Among them is the transportation of payloads to Low Earth Orbit (LEO) as well as the delivery of supplies to the International Space Station (ISS). These spaceports serve as launch facilities for different space agencies, private companies, and international collaborations, enabling the exploration and utilization of space for scientific, commercial, and technological advancements.[6]

Space activities have a wide range of applications and benefits for humanity, encompassing both scientific research and commercial endeavors. Some of these applications include communications, Earth observation, meteorology, and security. Each of these space utilization areas require satellites positioned in different orbits depending on their mission. One of the key orbits utilized for various purposes is LEO. LEO is particularly advantageous for applications such as Earth observation and satellite constellations, as it provides closer proximity to the Earth's surface, enabling higher-resolution imaging, faster data transmission, and shorter communication delays. Additionally, LEO is commonly used for missions involving the ISS and the deployment of small satellites for a wide range of purposes, including telecommunications, scientific research, and environmental monitoring.

The LEO, according to the Inter-Agency Space Debris Coordination Committee, is a concentric region extending up to 2,000 km from the Earth's surface.[7] The abundance of satellites in LEO is driven by its advantages, such as closer proximity to the Earth, which enables higher-resolution imaging, faster data transmission, and lower communication latency. Additionally, LEO offers better coverage and capacity for communication services, including broadband internet connectivity, as well as efficient Earth observation capabilities for monitoring weather patterns, environmental changes, and other remote sensing applications.

The growing number of satellites in LEO and the expanding chain of value associated with this orbit highlight its significance in enabling numerous technological advancements and services that benefit various industries and aspects

[5] Legislative Information Secretariat of the Ministry of the Interior, (2020), Initiative that reforms Articles 28 and 73 of the Political Constitution of the United Mexican States, http://sil.gobernacion.gob.mx/Archivos/Documentos/2020/10/asun_4091344_20201014_1602716063.pdf.

[6] Center for Strategic & International Studies, (2019), "*Spaceports of the World*", https://csis-website-prod.s3.amazonaws.com/s3fs-public/publication/190314_Spaceports_The_World.pdf.

[7] IADC, (2007), "*IADC Space Debris Mitigation Guidelines*", www.unoosa.org/documents/pdf/spacelaw/sd/IADC-2002-01-IADC-Space_Debris-Guidelines-Revision1.pdf.

of human life. According to the Space Foundation, the space economy reached approximately U.S. $469 billion in 2021.[8] This economy can be segmented into three types of activities: (1) upstream activities directly related to research and development of space technology, (2) downstream activities that depend on space technology, and (3) activities derived from space.

As per the Federal Aviation Administration (FAA)[9] of the USA, spaceports are launch and reentry sites for launch vehicles. This work will focus on vertical launches with rocket vehicles. These ports are crucial for defining the rocket's flight trajectory because, to position a satellite in orbit, it is necessary to first elevate it to the desired height above the Earth's surface and achieve sufficient escape velocity to keep the object orbiting at that specific altitude.

Such launch sites have a significant impact on the economy of the region and the nation at large due to the accompanying needs, such as operations, production, and transportation of propellants and oxidizers, which are considered part of the downstream industry. Furthermore, it directly influences the employment opportunities generated through the construction, operation, and even visits to the spaceport.

According to the National Space Activities Program 2020–2024 of the Mexican Space Agency (AEM), in order to achieve objective number 3, "Increase capabilities and promote cooperation in the country's science and technology, in space exploration for the scientific and technological strengthening of Mexico," the prioritized strategy 3.1 is proposed: "Promote the development of indigenous capabilities to achieve technological independence in space exploration for the benefit of Mexicans." This strategy, in turn, requires a specific action 3.1.4: "Establish collaboration programs through alliances with higher education institutions, research centers, and the private sector to promote rocket design and development activities."[10]

Currently, one of the actions required by the AEM is to promote the development of rockets. In Mexico, there are various rocketry teams and clubs, the majority of which belong to universities, with the rest being actors from the private sector. Due to the growing demand for manufacturing their own space rocket engines, it is necessary to encourage government authorities to support the necessary reforms to existing laws and create regulations that promote the secure use of rocket engines, as well as the chemical compounds required for their operation. Initially, this would involve suborbital amateur flights, thus laying the groundwork for the utilization of these compounds for possible orbital missions.

[8] Space Foundation, (2022), "*The Space Report Q2*", https://www.spacefoundation.org/2022/07/27/the-space-report-2022-q2/.

[9] FAA, (2022), "*Spaceports*", https://www.faa.gov/space/spaceports.

[10] Official Gazette of the Federation, 01, 63–207. "*National Space Activities Program 2024*", Mexican Space Agency, https://www.dof.gob.mx/nota_detalle.php?codigo=5608451&fecha=22/12/2020#gsc.tab=0.

3 Orbital Spaceflight

Establishing an indigenous Mexican spaceport can be a great opportunity to boost space activities in Mexico, as well as in different segments of the value chain, which in turn could generate greater economic development in the country. The Mexican aerospace sector is currently focused on the development of CubeSat technologies,[11] but the exponential growth of skilled human capital opens the door to research and development of more aerospace projects. This provides the possibility of venturing into launch vehicles, positioning Mexico in aerospace technology development and directly benefiting the economy, academia, industry, and government. The geographical expanse of Mexico also presents an excellent opportunity to venture into launch platform infrastructure for rocket vehicles. Together, these opportunities can lead to a significant increase in space activity in Mexico and strengthen its position in the global aerospace industry.

The mission of a launch vehicle is segmented into three different flight stages: (1) Launch Phase, (2) Cruise Phase and (3) Encounter Phase.[12] The focus will be on the Launch Phase to define the parameters that influence it and use them as requirements for the geographical location of a launch site in Mexican territory.

3.1 Orbital Elements

It is important to consider classical orbital elements such as the inclination of the orbital plane respect to the equatorial plane (i), eccentricity (e), argument of perigee (ω), true anomaly (θ), ascending and descending nodes, and the right ascension of the ascending node (Ω). The following graphics show the mentioned elements (Fig. 1).

Previously, it was mentioned that in order to put an object into orbit, it is necessary to reach a enough altitude above the surface of the Earth and have a speed greater than the escape velocity so that the satellite does not fall into freefall as a result of gravity and atmospheric drag.

Firstly, to reach that altitude, a enough amount of fuel is needed to generate the thrust that lifts the rocket to that position. The calculations to determine the amount of fuel depend on the propellant and oxidizer required, as well as the performance of the engine, the thrust it generates, the required escape velocity and a term called Delta V-Budget. Nevertheless, those concepts are beyond the scope of this work.

[11] ProMéxico, AEM, (2017), "*Plan de Órbita 2.0*", www.gob.mx/cms/uploads/attachment/file/414932/Plan_Orbita_2.0.pdf.

[12] Kluever C., (2018), "Space Flight Dynamics", Launch Trajectories, p. 219.

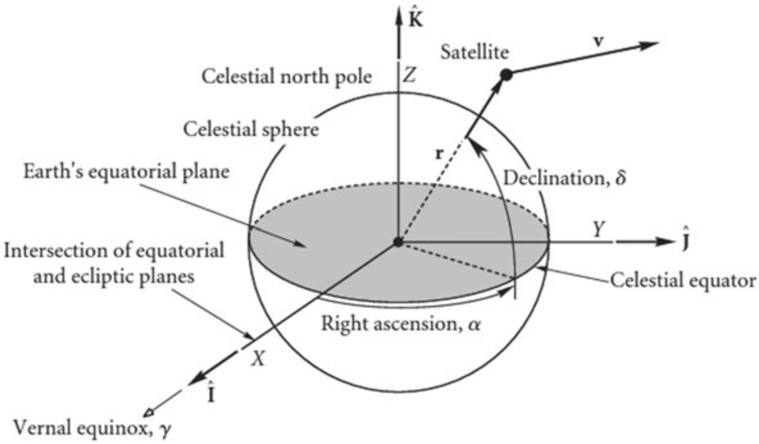

Fig. 1 **Angular Orbital Elements.** Curtis H. (2005). "Orbital mechanics for students" Fig. 4.5 Geocentric Equatorial Frame, p.155

3.2 Escape Velocity

On the other hand, the escape velocity depends on the altitude one wishes to reach and the mass of the body generating the gravitational attraction (in this case, the Earth). The equation is expressed as.

$$v_{esc} = \sqrt{\frac{2\mu}{r}} \tag{1}$$

where r is the radius of the orbit or the distance from Earth center to the satellite. And $\mu = GM$ is the geocentric gravitational constant; G is the gravitational constant of Earth and M is its mass.[13]

- Earth Radius: $R_e = 6{,}0.378[km]$
- Standard Gravitational parameter $GM = 3{,}986 \times 10^{14}[\frac{m^3}{s^2}]$
- Altitude $h = 160[km]$
- Orbit radius $R + h = r = 6{,}0.538[km]$

In this case, a circular orbit of 160 km in altitude or 6,0.538 km in radius with respect to the centre of our planet is proposed, for which the respective unit conversions and substitution of the values in Eq. 1 gives an escape velocity of 11,042 [km/s]. To reach this velocity, the Earth's rotational speed of 465 [m/s] can be taken advantage of, and to make the most of it, the launch should be as close

[13] Ibid., 39.

Fig. 2 Proposed coordinates in Baja California Sur (Google earth, "Vista aérea del El Conejo" [Satellite image])

as possible to the equator, since the velocity varies according to latitude, and the launch direction should preferably be eastwards. The approximate velocity in a given area can be calculated using the following equation:

$$v = v_a * \cos(l) \tag{2}$$

where v is the velocity we want to obtain, v_a is the speed of the Earth at the equator and l is the latitude of the area in question.[14]

3.3 Case Study

The following is a case study proposing a launch location at coordinates 24,072,576,097,343,966 NORTH, −111,0,057,264,686,843,027 WEST, located in Baja California Sur (Fig. 2).

[14] Tewari A., (2007), "*Atmospheric and Spaceflight Dynamics*", Translational Motion, p. 62.

Thus, the launch site is adjacent to the Atlantic Ocean for stage release and the proposed orbit is retrograde. It goes in the opposite direction to the Earth's rotational direction in order not to fly overpopulated areas of its own country and not to intervene in the airspace of another nation.

The angle of inclination and azimuth will be proposed by the calculations of the following equation for which the azimuth is defined as the latitude of the launch site.

$$\varphi_1 = \arcsin(\frac{cos\theta}{cos\beta}) \tag{3}$$

By substituting values from 170° to 250° for the angle of inclination i, we can obtain the following results for azimuth. Since it is a retrograde orbit, we obtain the inclination angles by the following equation (see Table 1).[15]

$$\theta = 180 - i \tag{4}$$

3.4 Geographical Considerations

Geographical considerations include climate, seismicity, accessibility to roads and proximity to towns and natural protected areas. According to data from the National Commission for the Knowledge and Use of Biodiversity (CONABIO), the chosen region has a very dry semi-warm climate with rainfall in summer, its average annual temperature is 14-24ºC, while the total annual rainfall is less than 100 mm.[16]

It is important to consider the presence of possible cyclones and hurricanes. As shown in Fig. 3, compared to other locations in Mexico, Baja California Sur presents on average a lower number of hurricanes greater than category 3 on the Saffir-Simpson scale, which classifies tropical cyclones according to wind intensity.[17]

Currently, one of the actions required by the AEM is to promote the development of national rockets. In Mexico, there are various rocketry teams and clubs, the majority of which belong to universities, with the rest being actors from the private sector. Due to the growing demand for manufacturing their own space rocket engines, it is necessary to encourage government authorities to support the necessary reforms to existing laws and create regulations that promote the use and utilization of rocket engines, as well as the chemical compounds required for their

[15] Walter U., (2018), "*Astronautics. The Physics of Space Flight*", 8.6.1 Launch Phase, p. 356.

[16] CONABIO, (w.d.), Sierra de la Laguna, www.conabio.gob.mx/conocimiento/regionalizacion/doctos/rtp_001.pdf.

[17] National Hurricane Center, (n.d.), History of the National Hurricane Center, www.nhc.noaa.gov/outreach/history/.

Table 1 Azimuth versus Retrograde inclination angle

Retrograde inclination angle (°)	Azimuth
35	85.185
36	79.783
37	76.291
38	73.453
39	70.974
40	68.728
41	66.647
42	64.69
43	62.831
44	61.051
45	59.336
46	57.675
47	56.061
48	54.486
49	52.947
50	51.438
51	49.956
52	48.498
53	47.062
54	45.645
55	44.246
56	42.862
57	41.494
58	40.138
59	38.794
60	37.462
61	36.14
62	34.827
63	33.522
64	32.226
65	30.937
66	29.655
67	28.38
68	27.11
69	25.845
70	24.586

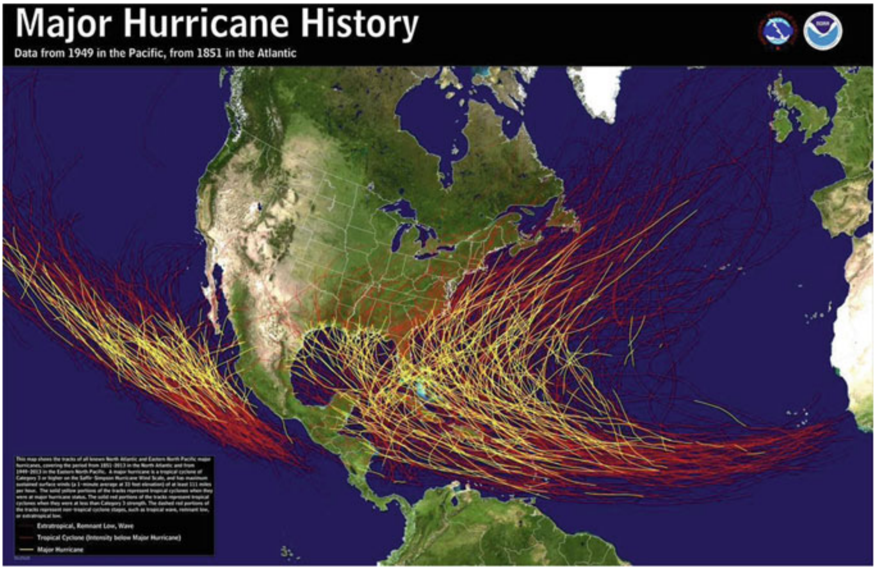

Fig. 3 **History of hurricanes from 1949 to 2013 in the Pacific Ocean.** National hurricane center. (n.d.). History of the national hurricane center. [Image]. https://www.nhc.noaa.gov/outreach/history/

operation. Initially, this would involve suborbital amateur flights, thus laying the groundwork for the utilization of these compounds for possible orbital missions.

4 Propellants

A propellant is the chemical mixture burned to produce thrust in rockets and consists of a fuel and an oxidizer. The fuel is a substance that burns when combined with an oxidizer for propulsion, while the oxidizer is an agent that releases oxygen to combine with a fuel. Propellants are classified based on their state: liquid, solid, or hybrid. In rocket engines, there are various types of propellants depending on the type of engine being used. A propellant therefore is composed of an oxidizer and a fuel.

The lists below do not include other propellants used in rocket engines. Their use depends on the type of fuel required for the rocket engine. Some of these substances are included in the General Pyrotechnics Law[18]:

[18] House of Representatives of the Honorable Congress of the Union, (n.d.), Federal Law on Pyrotechnics, www.diputados.gob.mx/servicios/datorele/cmprtvs/iniciativas/Inic/339/2.html.

Oxidizers: Potassium chlorate; Barium chlorate; Sodium chlorate; Strontium chlorate; Potassium perchlorate; Ammonium perchlorate; Barium nitrate; Strontium nitrate; Potassium nitrate; Sodium nitrate.

Fuels: Magnesium and its alloys with more than 50% powder content; Aluminum powder with particle size smaller than 250 mesh; Amorphous red phosphorus; White or yellow phosphorus.

Liquids: Liquid oxygen; Nitric acid; Dinitrogen tetroxide; Liquid fluorine; Liquid hydrogen; Hydrogen peroxide; Ethylene oxide; Kerosene; Liquid natural gas or LNG; Liquid methane.

Solids: Potassium nitrate; Ammonium nitrate; Sodium nitrate; Ammonium perchlorate; Sucrose; Dextrose; Sorbitol; Epoxy; Iron oxide; Silicone; Black Powder.

These lists provide examples of propellant components and are not exhaustive. The specific propellant composition depends on the requirements and design of the rocket engine. It is important to follow safety regulations and guidelines when handling and using rocket propellants.

4.1 Transport and Storage

The rocket engines of launch vehicles contain high-energy pyrotechnic materials, which classify them as "dangerous materials". Currently, in order to transport finished products and their components, there are established procedures by the Federal Registry of Firearms and Explosives Control of the Secretariat of National Defense (SEDENA) and the Directorate of Communications and Transportation (SCT). These procedures apply to products covered by the general permit, such as explosives, pyrotechnic devices, chemical substances, weapons, cartridges, and their components.

The SEDENA provides permits regarding explosives and chemical substances for pyrotechnic devices. These permits are necessary to ensure the safe transportation and handling of these dangerous materials.

Regarding explosives, the SEDENA issues permits for their manufacture, storage, transportation, sale, and use. These permits are subject to strict regulations to ensure public safety and prevent accidents.

In the case of chemical substances used in pyrotechnic devices, the SEDENA also grants permits for their acquisition, storage, transportation, and use. These chemical substances are subject to specific regulations and must comply with established safety standards.

It is important to note that permits and regulations related to explosives and chemical substances for pyrotechnic devices are subject to changes and updated by the relevant authorities. Therefore, it is crucial to stay informed and comply with the current legal requirements when transporting or handling these materials. It is recommended to directly contact the SEDENA for updated and detailed information regarding the specific permits required.[19]

These permits can be obtained online; however, a permit for engaging in rocket activities is excluded. Consequently, transportation companies do not include the transportation of commercial rocket motors. This is why having a formal regulation that encompasses precise regulations for packaging, labeling, storing, and transporting all forms of hazardous materials allows transportation companies to ship these engines by complying with these rules, just as some shipping companies do in other countries.

5 Launchment

Currently, the Civil Aviation Law in Mexico does not mention launch vehicles as objects that transit Mexican airspace, so activities related to their design and manufacturing of launch vehicles need to be regulated. In the Draft Official Mexican Standard PROY-NOM-080-SCT3-2000, which establishes the conditions for the practice of air sports, aerodesign is mentioned as a sport; however, rocketry is omitted.

In the United States of America, the Federal Aviation Administration (FAA) is the entity responsible for overseeing and regulating the use of rocket motors in the country. Amateur rocketry activities and commercial launches are regulated by the FAA.[20] Similarly, in Mexico, the AFAC (Federal Civil Aviation Agency) should establish a set of regulations regarding rocketry that can be used as a basis for launches into outer space.

In Title 14 "Aeronautics and Space" of the United States Code of Federal Regulations (CFR), the first requirement for any launch site is the owner's permission to use it for rocket flights.[21] The safety codes in the USA, such as those established by the National Association of Rockets (NAR) and the National Fire Protection Association (NFPA), set some minimum requirements for the size and surroundings of launch sites.

[19] Dirección General del Registro Federal de Armas de Fuego y Control de Explosivos, www.gob.mx/sedena/acciones-y-programas/formatos-de-pagos-e5-del-2023?state=published.

[20] FAA, (2023), Licenses, Permits and Approvals. www.faa.gov/space/licenses.

[21] USA Government, (2020), National Space Policy, https://trumpwhitehouse.archives.gov/wp-content/uploads/2020/12/National-Space-Policy.pdf.

5.1 Security

Currently in Mexico, there is the Federal Explosives Law based on NOM-008-STPS-1993.[22] However, there is no national Safety Code that helps establish safety protocols during experimentation with rocket motor propellants. It is necessary to handle oxidizers and fuels used in propellant formulation with care, and it is also important to regulate the materials used in rocket motor components such as the combustion chamber and nozzle.

In the USA, to regulate safety, there are the NFPA 1122 and 1127 standards regarding commercial motors, as well as rocketry associations that have established their own safety codes in the field of rocketry.[23]

The NFPA 1122 Rocket Model Code promotes safety through requirements for the design, construction, mass and power limitation, and reliability of model rocket motors and commercially produced model rocket motor reload kits and components for public educational, recreational, and competitive purposes.

The NFPA 1127 High-Power Rocket Code provides requirements for the safe operation of high-power rockets to protect users and the public from associated hazards that could cause death or injury. The application includes use in conjunction with education, recreation, and sporting competitions.[24]

The FAA has a set of specific regulations for rocket motors that apply to all stages of the life cycle of these devices, from manufacturing to launch and disposal. These regulations establish technical, operational, and safety requirements that must be met to ensure the safe and responsible use of rocket motors.

6 Regulation Proposals

Through the review of existing laws related to the use of explosives and pyrotechnic devices, it is necessary to make reforms to the current laws of Mexico.

- Accessible form by the AFAC for the Launch Authorization Request, where permission is requested and information is provided about the planned operation, operational area (location, altitudes, and direction), operating altitude, start and end dates and times of the launch.
- Safety code based on the Official Mexican Standard NOM-008-STPS-1993, issued by the Ministry of Labor and Social Welfare, regarding safety and hygiene conditions for the production, storage, and handling of explosives in

[22] Secretaria de la Defensa Nacional, (n.d.), Ley Federal de Pirotecnia, www.diputados.gob.mx/servicios/datorele/cmprtvs/iniciativas/Inic/339/2.htm.

[23] Tripoli Rocketry Association Safety Code, (2022), TAR, www.tripoli.org/content.aspx?page_id=22&club_id=795696&module_id=520420.

[24] National Fire Protection Association, (n.d.), NFPA 1127: Code for the management of high-power rocket fuels, www.nfpa.org/codes-and-standards/all-codes-and-standards/list-of-codes-and-standards/detail?code=1127.

the workplace, ensuring the integrity of experimenters and the general population by preventing accidents and incidents. This code should also consider the safety codes currently established by the National Association of Rocketry in the USA.

- Amendment to the Civil Aviation Law that includes launch vehicles as objects that traverse Mexican airspace, to promote the development of aerospace activities.
- Amendment to the pyrotechnics law that includes oxidizing and fuel substances used in rocket motors, making it easier to purchase, create, and transport them, as well as facilitating the purchase, transport, and storage of chemicals and finished propellants for experimentation.
- Space regulations developed by the Federal AFAC and the AEM to establish standards and rules governing suborbital launches as a first step, and subsequently, to establish regulations for orbital flights.

7 Conclusions

The establishment of a launch complex in Mexican territory could significantly drive the growth of the country's space industry and foster its economic development. Based on the calculations and results obtained, it is evident that the proposed area is a possibly good option for launches to LEO as it takes advantage of the Earth's rotational angular velocity, its proximity to the sea, and its distance from populated areas. Moreover, this site provides a wide range for launching into orbits with different inclinations, thereby increasing the possibility of accessing various orbits and expanding the types of missions that can be carried out.

It is necessary to implement safety protocols during the manufacturing and testing of a rocket engine, as there is currently no official standard in Mexico that regulates the handling of oxidizing and fuel substances in different types of rocket engines, firstly prioritizing model rocket design and development in order to reach orbital technology. It is important to create a safety protocol to prevent accidents and incidents related to propulsion activities of launch vehicles, whether they are experimental sounding rockets or commercial launchers. As well as the implementation of an official standard related to the propulsion systems of launch vehicles and an amendment to the current Pyrotechnics Law that includes the chemicals used in solid and liquid propellants of the engines is essential.

Finally, it is crucial to update and reform the legal frameworks existing in Mexico and propose a safety code established either by the government or space organizations established in Mexico. The main objective of regulating all space activities in a legal manner is to promote technological projects, thus achieving aerospace technological sovereignty and fostering the development of space activities in Mexico.

Hasel Ramírez Cortés is a Mexican space enthusiast currently pursuing her bachelor 's degree in Aeronautical Engineering at the National Polytechnic Institute. She has been involved in university engineering projects such as aerodesign and high-power rocketry. She has led the structural and aerodynamics design of sounding rockets for national and international rocketry competitions. She has collaborated in the structural design of solid rocket motors. She has also conducted research on structures and new materials for small satellites presented at Latin American and Caribbean Space Network and International Astronautical Congress. Currently, she is working on composite and smart materials for aerospace applications.

Angel Josafat Vázquez Minor is an aerospace engineering student and computer technician at National Autonomous University of Mexico. He is interested in jet propulsion, entrepreneurship, mechanical design, and disruptive technologies such as IoT, machine and deep learning. He has presented research projects at Latin American and Caribbean Space Network and International Astronautical Congress, representing the Mexican Space Agency. Currently, he leads rocketry, aerodesign and cubesat teams as well as research about hyperspectral remote sensing and satellite constellations. He is also a web developer in the Mexican Experimental Rocketry Meeting.